

RHCSA Examen Practice Tasks

1. Add 3 users: harry, natasha, tom. The requirements: The Additional group of the two users: harry, Natasha is the admin group. The user: tom's login shell should be non-interactive.

```
[root@workstation student]# groupadd admin
[root@workstation student]# useradd -G admin harry
[root@workstation student]# useradd -G admin Natasha
[root@workstation student]# useradd -s /sbin/nologin tom
[root@workstation student]# id harry
uid=1002(harry) gid=1003(harry) groups=1003(harry),1002(admin)
[root@workstation student]# id natasha
id: 'natasha': no such user
[root@workstation student]# id Natasha
uid=1003(Natasha) gid=1004(Natasha) groups=1004(Natasha),1002(admin)
[root@workstation student]# cat /etc/passwd
```

2. Plan to run echo hello command at 14:23 every day.

```
[root@workstation student]# crontab -e
```

```
23 14 * * * /bin/echo Hello
```

in Crontab

```
crontab: installing new crontab
```

```
[root@workstation student]# crontab -l
```

```
23 14 * * * /bin/echo Hello
```

verification

3. Create a user named ana, and the user id should be 1244, and the password should be alex111.

```
[root@workstation student]# useradd -u 1244 ana
```

```
[root@workstation student]# passwd ana
```

```
Changing password for user ana.
```

4. Set the hostname for your machine to be host@example.com

```
[root@workstation student]# nmtui
```

Set system hostname

5. Securely copy /root/shared files from your machine to "server2.example.com" to /shared directory, ssh connection is not established, use the password "password"

```
[root@workstation student]# scp /root/shared server.example.com:/shared/
```

6. Configure the IP 192.168.122.10 on eth0 interface on 'system.eight.example.com' and set the DNS IP as 192.168.122.254

```
[root@workstation student]# nmtui
```

Edit a connection

IPv4 CONFIGURATION		<Manual>	
Addresses	172.25.250.21/24	<Remove>	
	<Add...>		
Gateway	172.25.250.254	<Remove>	
DNS servers	172.25.250.254	<Remove>	
	<Add...>		
Search domains	lab.example.com	<Remove>	
	example.com	<Remove>	

Remember to deactivate□ Reactivate connection so new conf will be loaded

7. Configure the Default Gateway as 192.168.122.1

```
[root@workstation student]# nmtui
```

Edit a connection

IPv4 CONFIGURATION		<Manual>	
Addresses	172.25.250.21/24	<Remove>	
	<Add...>		
Gateway	172.25.250.254	<Remove>	
DNS servers	172.25.250.254	<Remove>	
	<Add...>		
Search domains	lab.example.com	<Remove>	
	example.com	<Remove>	

Remember to deactivate□ Reactivate connection so new conf will be loaded

8. Copy all the lines starting with "ted" "Ted" from ted.txt and copy it to file onlyted.txt

```
[root@workstation student]# cat ted.txt | grep -i ^ted > /onlyted.txt
[root@workstation student]# cat /onlyted.txt
ted
teddy
Tedros
```

9. Schedule a script /myawesomescript.sh as user "nick" which should be executed 12:15 every Tuesday.

```
[root@workstation student]# useradd nick
[root@workstation student]# vi /etc/cron.allow
[root@workstation student]# su nick
```

In cron.allow just write the name of the user who is allowed to access crontab in this case "nick"

```
[nick@workstation student]$ crontab -e
no crontab for nick - using an empty one
crontab: installing new crontab
[nick@workstation student]$ crontab -l
15 12 * * 2 /myawesomescript.sh
[nick@workstation student]$
```

10. Configure administrative rights for all the members of the testers group

```
[root@workstation student]# groupadd testers
[root@workstation student]# echo "%testers ALL=(ALL) ALL" > /etc/sudoers.d/teste
rs
[root@workstation student]# cat /etc/sudoers.d/testers
%testers ALL=(ALL) ALL
[root@workstation student]#
```

11. Force all the testers to change their password whenever they log on to the system for the first time

```
[root@workstation student]# chage -d 0 tester1
[root@workstation student]# chage -d 0 tester2
[root@workstation student]# chage -d 0 tester3
[root@workstation student]#
```

12. Add group called testers with GID 30000

```
[root@workstation student]# groupmod -g 30000 testers
```

If group does not exist!

```
[root@workstation student]# groupadd -g 30000 testers
```

13. Change the default password policy so that newly created users have passwords that must be changed every 60 days

```
[root@workstation student]# vi /etc/login.defs
```

☐ Default user creation login configuration file

```
PASS_MAX_DAYS 60
PASS_MIN_DAYS 0
PASS_MIN_LEN 5
PASS_WARN_AGE 7
```

☐ Change max PASS_MAX_DAYS to 60 or whatever is required

14. Change the password policy for tester3 to require a new password every 10 days

```
[root@workstation student]# chage -M 10 tester3
```

15. Set the users account for tester1, tester2, tester3 to expire in 30 days

```
[root@workstation student]# date -d "+30 days" +%F
2021-01-17
[root@workstation student]# chage -E 2021-01-17 tester1
[root@workstation student]# chage -E 2021-01-17 tester2
[root@workstation student]# chage -E 2021-01-17 tester3
```

```
[root@workstation student]# chage -l tester1
```

 Validation

First Determine date as required then set expiry date through the use of chage -E

16. Configure the hosts file so that 10.0.0.5 could be referenced as **private**

```
[root@workstation student]# echo -e "10.0.0.5 private" >> /etc/hosts
```

Or modify hosts file in /etc/hosts with the same text using vi

17. Configure your system to use the repos

at <http://repo.eight.example.com/BaseOS> <http://repo.eight.example.com/AppStream> (GPG Check On)

Create one file for repo you would like to add (DO NOT FORGET THEY HAVE TO END IN .REPO!!!!)

```
[root@workstation student]# vi /etc/yum.repos.d/Base0s.repo
```

```
[root@workstation student]# vi /etc/yum.repos.d/Appstream.repo
```

Each file should contain this information depending on the requirements

```
[Appstream]
name=CHOOSENAME
baseurl=URL
enabled=1
gpgcheck=1
```

18. The server is re-located to Bahamas, configure the timezone appropriately.

```
[root@workstation student]# tzselect
Please identify a location so that time zone rules can be set correctly.
Please select a continent, ocean, "coord", or "TZ".
1) Africa
2) Americas
3) Antarctica
4) Asia
5) Atlantic Ocean
6) Australia
7) Europe
8) Indian Ocean
9) Pacific Ocean
10) coord - I want to use geographical coordinates.
11) TZ - I want to specify the time zone using the Posix TZ format.
```

Use either tzselect or timedatectl commands the rest should be easy to determine

19. Configure the chronyd service on server1 to synchronize the system time with the NTP time source at repo.eight.example.com

Timedatectl to verify

Timedatectl set-ntp yes to turn on NTP sync

```
[root@workstation student]# timedatectl
      Local time: Fri 2020-12-18 05:51:11 EST
      Universal time: Fri 2020-12-18 10:51:11 UTC
      RTC time: Fri 2020-12-18 10:51:10
      Time zone: America/New_York (EST, -0500)
System clock synchronized: yes
      NTP service: active
      RTC in local TZ: no
[root@workstation student]# timedate set-ntp yes
bash: timedate: command not found...
[root@workstation student]# timedatectl set-ntp yes
```

Edit chrony.conf to add the NTP Server

```
[root@workstation student]# vi /etc/chrony.conf
```

Add: server repo.eight.example.com to the /etc/chrony.conf

20. Create symbolic link for /home/vagrant/test that will be named /tmp/test-link

```
[root@workstation student]# touch /home/vagrant/test
[root@workstation student]# ln -s /home/vagrant/test /tmp/test-link

[root@workstation student]# ls -al /tmp/test-link
lrwxrwxrwx. 1 root root 18 Dec 18 05:59 /tmp/test-link -> /home/vagrant/test
```

21. Create hard link for /home/vagrant/test that will be named /tmp/test-hard

```
[root@workstation student]# ln /home/vagrant/test /tmp/test-hard
```

22. Use tar command to archive all contents of /etc directory in /root/etc.tar file. Compress the archived files using gzip

-c Create

-v verbose

-t View content

-z gzip

-j bzip2

-J xz

```
[root@workstation student]# tar -czf /root/etc.tar.gz /etc
```

```
[root@workstation student]# tar -tzf /root/etc.tar.gz checking !
```

Part 2:

- Set the SELinux to enforcing mode.
- The root password is lost, interrupt the boot process and set the root password to "newpassword".
- Set the Graphical target to be the default one.

```
[root@workstation student]# systemctl set-default graphical.target
Removed /etc/systemd/system/default.target.
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/graphical.target.
[root@workstation student]#
```

- Find all the files owned by user "alex" and redirect the output to /home/alex/files
- Create a 2G swap partition which take effect automatically at boot-start, and it should not affect the original swap partition.
- Find the files owned by harry, and copy it to catalog: /opt/dir

- g. System1 should boot into the multiuser target

```
[root@workstation student]# systemctl set-default multi-user.target
Removed /etc/systemd/system/default.target.
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
[root@workstation student]#
```

- h. Create a volume group called shazam consists of two physical volumes each 256MB in size
- i. Reset Root Password (No solution to Show)
- j. Troubleshoot FSTAB (No solution to Show)

Check Disk and we shall use SDB

```
[root@server1 ~]# lsblk
```

NAME	MAJ:MIN	RM
sda	8:0	0
├─sda1	8:1	0
└─sda2	8:2	0
├─rhel_rhel8-root	253:0	0
└─rhel_rhel8-swap	253:1	0
sdb	8:16	0

We shall create 2 partitions

```
[root@server1 ~]# fdisk /dev/sdb
```

Welcome to fdisk (util-linux 2.32.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Command (m for help): n
Partition type
 p primary (1 primary, 1 extended, 2 free)
 l logical (numbered from 5)
Select (default p):

Using default response p.
Partition number (1,2, default 1):
First sector (2048-10533415, default 2048):
Last sector, +sectors or +size{K,M,G,T,P} (2048-1050623, default 1050623): +256M

Created a new partition 1 of type 'Linux' and of size 256 MiB.

Type LVM

```
Command (m for help): t
Partition number (1,3-5, default 5): 1
Hex code (type L to list all codes): 8e

Changed type of partition 'Linux' to 'Linux LVM'.
```

Create Physical Volumes

```
[root@server1 ~]# pvcreate /dev/sdb1 /dev/sdb2
Physical volume "/dev/sdb1" successfully created.
Physical volume "/dev/sdb2" successfully created.
```

Create Volume Group

```
[root@server1 ~]# vgcreate shazam /dev/sdb1 /dev/sdb2
Volume group "shazam" successfully created
```


Verify volume Group

```
[root@server1 ~]# vgdisplay shazam
--- Volume group ---
VG Name                shazam
System ID
Format                 lvm2
Metadata Areas         2
Metadata Sequence No   1
VG Access               read/write
VG Status               resizable
MAX LV                 0
Cur LV                 0
Open LV                0
Max PV                 0
Cur PV                 2
Act PV                 2
VG Size                 504.00 MiB
PE Size                 4.00 MiB
Total PE                126
Alloc PE / Size         0 / 0
Free PE / Size          126 / 504.00 MiB
VG UUID                 NPFWy6-1Gx7-ayve-eSYa-snTL-Rtyn-ey1anG
```

- k. Add a logical volume called storage in the volume group shazam mounted at /storage permanently as xfs

Create Logical Volume

```
[root@server1 ~]# lvcreate -n storage -L 400M shazam
Logical volume "storage" created.
```

Verify

```
[root@server1 ~]# lvs
```

LV	VG	Attr	LSize	Pool	Origin	Data%	Meta%	Move	Log	Cpy%	Sync	Conv
root	rhel_rhel8	-wi-ao----	28.93g									
swap	rhel_rhel8	-wi-ao----	2.07g									
storage	shazam	-wi-a-----	400.00m									

Format

```
[root@server1 ~]# mkfs -t xfs /dev/shazam/storage
```

meta-data=/dev/shazam/storage	isize=512	agcount=4, agsize=25600 blks
=	sectsz=512	attr=2, projid32bit=1
=	crc=1	finobt=1, sparse=1, rmapbt=0
=	reflink=1	
data	=	bsize=4096, blocks=102400, imaxpct=25
=	sunit=0	swidth=0 blks
naming	=version 2	bsize=4096, ascii-ci=0, ftype=1
log	=internal log	bsize=4096, blocks=1368, version=2
=	sectsz=512	sunit=0 blks, lazy-count=1
realtime	=none	extsz=4096, blocks=0, rtextents=0

Create Mountpoint

```
[root@server1 ~]# mkdir /storage
```

Enter fstab and make it persistent by adding

```
/dev/shazam/storage /storage xfs defaults 1 2
```

Test

```
[root@server1 ~]# mount -a
```

Verify LV

```
[root@server1 ~]# df -h
df: -: No such file or directory
df: h: No such file or directory
[root@server1 ~]# df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	901M	0	901M	0%	/dev
tmpfs	915M	0	915M	0%	/dev/shm
tmpfs	915M	25M	891M	3%	/run
tmpfs	915M	0	915M	0%	/sys/fs/cgroup
/dev/mapper/rhel_rhel8-root	29G	9.9G	20G	35%	/
/dev/sda1	1014M	161M	854M	16%	/boot
tmpfs	183M	0	183M	0%	/run/user/1000
/dev/mapper/shazam-storage	395M	24M	372M	6%	/storage

I. Extend the logical volume storage to be 700MB:

Create new partitions if you do not have space

```
Command (m for help): n
All primary partitions are in use.
Adding logical partition 5
First sector (2101248-10533415, default 2101248):
Last sector, +sectors or +size{K,M,G,T,P} (2101248-10533415, default 10533415): +512M

Created a new partition 5 of type 'Linux' and of size 512 MiB.

Command (m for help): t
Partition number (1-5, default 5):
Hex code (type L to list all codes): 8e

Changed type of partition 'Linux' to 'Linux LVM'.
```

CREATE PV

```
[root@server1 ~]# pvcreate /dev/sdb5
Physical volume "/dev/sdb5" successfully created.
```

EXTEND VG

```
[root@server1 ~]# vgextend shazam /dev/sdb5
Volume group "shazam" successfully extended
```

EXTEND LV

```
[root@server1 ~]# lvextend -L 700MB /dev/shazam/storage
Size of logical volume shazam/storage changed from 400.00 MiB (100 extents) to 700.00 MiB (175 extents).
Logical volume shazam/storage successfully resized.
```

EXTEND filesystem

```
[root@server1 ~]# xfs_growfs /storage
meta-data=/dev/mapper/shazam-storage isize=512    agcount=4, agsize=25600 blks
        =                               sectsz=512   attr=2, projid32bit=1
        =                               crc=1        finobt=1, sparse=1, rmapbt=0
        =                               reflink=1
data      =                               bsize=4096   blocks=102400, imaxpct=25
        =                               sunit=0      swidth=0 blks
naming    =version 2                     bsize=4096   ascii-ci=0, ftype=1
log        =internal log                 bsize=4096   blocks=1368, version=2
        =                               sectsz=512   sunit=0 blks, lazy-count=1
realtime  =none                          extsz=4096   blocks=0, rtextents=0
data blocks changed from 102400 to 179200
```

Verify

```
[root@server1 ~]# df -h /storage
Filesystem                Size  Used Avail Use% Mounted on
/dev/mapper/shazam-storage 695M   26M  670M   4% /storage
```

m. Set the SELinux to permissive mode.

CHECK ENFORCE

```
[root@server1 ~]# getenforce
```

SET ENFORCE by changing to permissive here with VI

```
[root@server1 ~]# vim /etc/selinux/config
```

n. Allow your web server httpd to access files at **/var/web-content**, Configure your SELinux to allow that.

CHECK CONTEXT

```
[root@server1 ~]# ls -ld -Z /var/www/html/
system_u:object_r:httpd_sys_content_t:s0 /var/www/html/
```

CREATE FOLDER

```
mkdir -p /var/web-content
```

COMPARE CONTEXTS

```
[root@server1 ~]# ls -d -Z /var/www/html/ /var/web-content/  
unconfined_u:object_r:var_t:s0 /var/web-content  
system_u:<b>object_r:httpd_sys_content_t</b>:s0 /
```

ASSIGN NEW CONTEXT

```
[root@server1 ~]# semanage fcontext -a -t httpd_sys_content_t '/var/web-  
content(/.*)?'
```

RESTORE LABELS

```
[root@server1 ~]# restorecon -R /var/web-content/
```

YOU CAN FIND A GOOD EXAMPLE IN THE: (man semanage fcontext) at the end

- o. Configure httpd to use port 1001

INSTALL SEMANAGE IF NOT PRESENT

```
[root@server1 ~]# yum install policycoreutils-python-utils -y
```

ADD PORT TO HTTPD

```
[root@server1 ~]# semanage port -a -t http_port_t -p tcp 1001
```

ADD PORT TO ZONE

```
[root@server1 ~]# firewall-cmd --get-default-zone  
public  
[root@server1 ~]# firewall-cmd --set-default-zone public  
[root@server1 ~]# firewall-cmd --permanent --zone=public --add-port=1001/tcp
```

- p. Create a thinly provisioned filesystem called **practice-fs** using stratis in a pool named **practice-pool** mounted at **/practice-stratis-volume**

INSTALL

```
[root@server1 ~]# yum install stratisd stratis-cli
```

ENABLE AND START

```
[root@server1 ~]# systemctl enable --now stratisd
```

CREATE POOL

```
[root@server1 ~]# stratis pool create practice-pool /dev/sdc
```

ADD ADDITIONAL DEVICES TO POOL

```
stratis pool add-data practice-pool /dev/sdb
```

VERIFY POOL

```
[root@server1 ~]# stratis pool list
```

Name	Total Physical Size	Total Physical Used
practice-pool	1 GiB	52 MiB

ADD Filesystem

```
[root@server1 ~]# stratis filesystem create practice-pool practice-fs
```

VERIFY FS

```
[root@server1 ~]# stratis filesystem list
```

Pool	Name	Name	Used	Created	Device	UUID
practice-pool	practice-fs	545 MiB	Mar 02 2020 10:04	/stratis/practice-pool/practice-fs	e9a9032800044f	

GET UUID

```
[root@server1 ~]# lsblk --output=UUID /stratis/practice-pool/practice-fs
UUID
e9a90328-0004-4fcb-98db-7517b7322649
```

CREATE MOUNTPOINT

```
[root@server1 ~]# mkdir /practice-stratis-volume
```

MAKE IT PERSISTENT BY ADDING IN FSTAB

```
UUID=e9a90328-0004-4fcb-98db-7517b7322649 /practice-stratis-volume xfs defaults,x-
systemd.requires=stratisd.service 0 0
```

TEST

```
mount /practice-stratis-volume
```

q. Add 512MB swap partition

ADD PARTITION


```
[root@server1 ~]# fdisk /dev/sdb

Welcome to fdisk (util-linux 2.32.1).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.


Command (m for help): n
Partition type
   p   primary (2 primary, 0 extended, 2 free)
   e   extended (container for logical partitions)
Select (default p): p
Partition number (3,4, default 3):
First sector (1050624-10533415, default 1050624):
Last sector, +sectors or +size{K,M,G,T,P} (1050624-10533415, default 10533415): +512M

Created a new partition 3 of type 'Linux' and of size 512 MiB.
```

CHANGE TYPE TO SWAP

```
Command (m for help): t
Partition number (1-3, default 3): 3
Hex code (type L to list all codes): 82

Changed type of partition 'Linux' to 'Linux swap / Solaris'.
```

FORMAT

```
[root@server1 ~]# mkswap /dev/sdb3
Setting up swapspace version 1, size = 512 MiB (536866816 bytes)
no label, UUID=a168f157-0232-4406-b676-5a0a66fc5e93
```

MAKE IT PERSISTENT

```
[root@server1 ~]# vi /etc/fstab
UUID=a168f157-0232-4406-b676-5a0a66fc5e93 swap swap defaults 0 0
```

TEST

```
[root@server1 ~]# swapon -a
```

VERIFY

```
[root@server1 ~]# swapon -s
```

Filename	Type	Size	Used	Priority
/dev/dm-1	partition	2170876	0	-2
/dev/sdb3	partition	524284	0	-3

- r. Create cdo volume practice-vdo with the device /dev/vdd, set the logical size to 50 GB

CREATE VDO

```
[root@server1 ~]# vdo create --name=practice-vdo --device=/dev/vdd --
vdoLogicalSize=50G
```

VERIFY

```
[root@server1 ~]# vdo list
```

- s. Mount the newly created vdo on **/vdo-practice** with **XFS** filesystem so that it persists across reboots.

FORMAT

```
[root@server1 ~]# mkfs.xfs -K /dev/mapper/practice-vdo
```

REGISTER NEW DEVICE

```
[root@server1 ~]# udevadm settle
```

MAKE MOUNTPOINT

```
[root@server1 ~]# mkdir /vdo-practice
```

DETERMINE UUID

```
[root@server1 ~]# lsblk --output=UUID /dev/mapper/practice-vdo
```

```
UUID
```

MAKE IT PERSISTENT BY ADDING IN /etc/FSTAB

```
UUID=ef8...b1 /vdo-practice xfs defaults,x-systemd.requires=vdo.service 0 0
```

MOUNT

```
[root@server1 ~]# mount /vdo-practice
```